Digits Identifier Circuit for Euler's Number

PC/CP220 Revised Project Phase I

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Description:

The project aims to design a circuit that identifies specific decimal digits of the mathematical constant e (approximately 2.71828) based on binary inputs representing decimal places (0-9). It will provide outputs corresponding to the binary representation of those digits.

Inputs:

The Digit Identifier Circuit will have four inputs:

1. **A0** (Least Significant Bit)
   - Input: 1 if included, 0 if not.
2. **A1**
   - Input: 1 if included, 0 if not.
3. **A2**
   - Input: 1 if included, 0 if not.
4. **A3** (Most Significant Bit)
   - Input: 1 if included, 0 if not.

Input Specification:

- The circuit will read the inputs as a 4-bit binary number (A3 A2 A1 A0), representing decimal places from 0 to 9.

Outputs:

The Digit Identifier Circuit will have at least ten outputs, each corresponding to a digit of e. The outputs will be:

1. **Digit 0** (O0)
2. **Digit 1** (O1)
3. **Digit 2** (O2)
4. **Digit 3** (O3)
5. **Digit 4** (O4)
6. **Digit 5** (O5)

7. **Digit 6** (O6)
8. **Digit 7** (O7)
9. **Digit 8** (O8)
10. **Digit 9** (O9)

Output Logic:

- An output will be set to ‘1’ if it corresponds to the digit of e indicated by the binary input; otherwise, it will be set to ‘0’.
- The digits of e are:
  - 2.7182818284 → (0: `2`, 1: `7`, 2: `1`, 3: `8`, 4: `2`, 5: `8`, 6: `1`, 7: `8`, 8: `4`, 9: `0`)
- Example outputs:
  - Input `0000` (decimal `0`) → Output `10` (binary for digit `2`)
  - Input `0001` (decimal `1`) → Output `111` (binary for digit `7`)
  - Input `0010` (decimal `2`) → Output `1` (binary for digit `1`)
  - Input `0011` (decimal `3`) → Output `1000` (binary for digit `8`), etc.

Handling Combinations:

- For each binary input, only one output will be set to 1, corresponding to the digit of e for the specified decimal place.
- If the input is outside the valid range (e.g., 1010 or greater), all outputs will be set to 0 as no valid digit exists for that input.

Error Conditions and Responses:

Invalid Input Range:

- Condition: When the binary input exceeds 0001 1001 (decimal 9), indicating an invalid request for digits beyond the 10th decimal place of ee.
- Response:
  - Set all outputs (O0 to O9) to 0..

Non-binary Inputs:

- Condition: If any input bit is not 0 or 1.
- Response:
    - Set all outputs (O0 to O9) to 0.

Ambiguous Possibilities:

Don't Care Conditions:

- Condition: There are no ambiguous possibilities in this design as each binary input uniquely corresponds to a single digit of e.
- Response:
    - If an input corresponds to a digit of e (within the range of 0-9), it will output the appropriate value.
    - If the input is outside the valid range or invalid (non-binary), all outputs will remain 0, providing a clear indication of the error.

Multiple Outputs:

- Condition: Each input corresponds to exactly one output. There are no cases where multiple digits would activate simultaneously based on valid inputs.
- Response:
    - Ensure that for valid inputs, only the respective output is set to 1, while all other outputs remain 0. This is to prevent confusion over multiple digits being identified for the same input.

Method:

Recommended Approach: Rounding

Reasoning:

1. **Precision Requirement**:
   - The goal of your circuit is to identify specific decimal digits of e. Since e has a non-terminating, non-repeating decimal expansion (approximately 2.7182818284), rounding is preferable to ensure that you accurately represent each digit.
2. **Significant Digits**:
   - Since the digits after the decimal point are significant in many mathematical and engineering contexts, rounding helps provide a more accurate representation of those digits when needed.
3. **Output Representation**:
   - For inputs representing decimal places, rounding ensures that if the digit of e falls exactly at a point where rounding decisions could be made (for example, at `2.718`), the circuit outputs the nearest digit correctly.

Conclusion:

Digit Identifier Circuit for Euler's number e it is intended to provide precise identification of the specific decimal digits in dependence on the 4-bit binary input that codes decimal places ranging from 0 to 9. Every single eligible input directly relates to a certain output associated with the digit of an integer. Thereby reducing the level of obscurity to an insignificant level. The circuit also takes precautions to handle an error condition involving an invalid input – in this case, all of the outputs are zeroed. By implementing rounding for output representation, the circuit improves on the accuracy of detecting meaning digits of e. In conclusion, this project shows an adequate appreciation of digital design concepts and is satisfactory from the technical point of view in meeting digit identification objectives.

Digits Identifier Circuit for Euler's Number

PC/CP220 Project Phase II

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Truth Table:

A3	A2	A1	A0	Decimal place	**e digit**	O0	O1	O2	O3	O4	O5	O6	O7	O8	O9
0	0	0	0	0	**2**	1	0	0	0	0	0	0	0	0	0
0	0	0	1	1	**7**	0	0	0	0	0	0	0	1	0	0
0	0	1	0	2	**1**	0	1	0	0	0	0	0	0	0	0
0	0	1	1	3	**8**	0	0	0	0	0	0	0	0	1	0
0	1	0	0	4	**2**	1	0	0	0	0	0	0	0	0	0
0	1	0	1	5	**8**	0	0	0	0	0	0	0	0	1	0
0	1	1	0	6	**1**	0	1	0	0	0	0	0	0	0	0
0	1	1	1	7	**8**	0	0	0	0	0	0	0	0	1	0
1	0	0	0	8	**2**	1	0	0	0	0	0	0	0	0	0
1	0	0	1	9	**8**	0	00	0	0	0	0	0	0	1	0

Logic Equation:

1. O_0 (Digit 2 at positions 0, 4, and 8)

   $O_0 = A3' \cdot A2' \cdot A1' \cdot A0' + A3' \cdot A2 \cdot A1' \cdot A0' + A3 \cdot A2' \cdot A1' \cdot A0'$
2. O_1 (Digit 1 at positions 2 and 6):

   $O_1 = A3' \cdot A2' \cdot A1 \cdot A0' + A3' \cdot A2 \cdot A1 \cdot A0'$
3. O_2 (No positions correspond to Digit 2 specifically, so O_2 is always 0):

   $O_2 = 0$
4. O_3 (No positions correspond to Digit 3 specifically, so O_3 is always 0):

   $O_3 = 0$
5. O_4 (No positions correspond to Digit 4 specifically, so O_4 is always 0):

   $O_4 = 0$
6. O_5 (No positions correspond to Digit 5 specifically, so O_5 is always 0):

   $O_5 = 0$
7. O_6 (No positions correspond to Digit 6 specifically, so O_6 is always 0):

   $O_6 = 0$
8. O_7 (Digit 7 at position 1):

   $O_7 = A3' \cdot A2' \cdot A1' \cdot A0$

9. O_8 (Digit 8 at positions 3, 5, 7, and 9):

   $O_8 =$
   $A3' \cdot A2' \cdot A1 \cdot A0 + A3 \cdot 'A2 \cdot A1' \cdot A0 + A3' \cdot A2 \cdot A1 \cdot A0 + A3 \cdot A2' \cdot A1' \cdot A0$

10. O_9 (No positions correspond to Digit 5 specifically, so O_9 is always 0):

    $O_9 = 0$

Testing Equation:

Input:

```
t1:not(a3) and not(a2) and not(a1) and not(a0) or not(a3) and a2 and not(a1) and not(a0) or a3 and not(a2) and not(a1) and not(A0);
t2:not(a3) and not(a2) and a1 and not(a0) or not(a3) and a2 and a1 and not a0;
t3:not(a3) and not(a2) and not(a1)and a0;
t4:not(a3) and not(a2) and a1 and a0 or a3 and not(a2) and not(a1) and a0 or not(a3) and a2 and a1 and a0 or a3 and not(a2) and
not(a1) and a0;

euler:t1 or t2 or t3 or t4;

euler,a0=false,a1=false,a2=false,a3=false;
euler,a0=true,a1=false,a2=false,a3=false;
euler,a0=false,a1=true,a2=false,a3=false;
euler,a0=true,a1=true,a2=false,a3=false;
euler,a0=false,a1=false,a2=true,a3=false;
euler,a0=true,a1=false,a2=true,a3=false;
euler,a0=false,a1=true,a2=true,a3=false;
euler,a0=true,a1=true,a2=true,a3=false;
euler,a0=false,a1=false,a2=false,a3=true;
euler,a0=true,a1=false,a2=false,a3=true;
```

Output:

$$\neg a_3 \land \neg a_2 \land \neg a_1 \land \neg a_0 \lor \neg a_3 \land a_2 \land \neg a_1 \land \neg a_0 \lor a_3 \land \neg a_2 \land \neg a_1 \land \neg A_0$$

$$\neg a_3 \land \neg a_2 \land a_1 \land \neg a_0 \lor \neg a_3 \land a_2 \land a_1 \land \neg a_0$$

$$\neg a_3 \land \neg a_2 \land \neg a_1 \land a_0$$

$$\neg a_3 \land \neg a_2 \land a_1 \land a_0 \lor a_3 \land \neg a_2 \land \neg a_1 \land a_0 \lor \neg a_3 \land a_2 \land a_1 \land a_0 \lor a_3 \land \neg a_2 \land \neg a_1 \land a_0$$

$$\neg a_3 \land \neg a_2 \land \neg a_1 \land \neg a_0 \lor \neg a_3 \land a_2 \land \neg a_1 \land \neg a_0 \lor a_3 \land \neg a_2 \land \neg a_1 \land \neg A_0 \lor \neg a_3 \land \neg a_2 \land a_1 \land \neg a_0 \lor \neg a_3 \land a_2 \land a_1 \land \neg a_0 \lor \neg a_3 \land \neg a_2 \land \neg a_1 \land a_0 \lor \neg a_3 \land \neg a_2 \land a_1 \land a_0 \lor a_3 \land \neg a_2 \land \neg a_1 \land a_0 \lor \neg a_3 \land a_2 \land a_1 \land a_0 \lor a_3 \land \neg a_2 \land \neg a_1 \land a_0$$

true

true

true

true

true

false

true

true

$\neg A_0$

true

Summary:

The Digit Identifier Circuit decodes certain decimal figures of Euler's number e. In other words, for operation we are using the 4-bit binary input, wherein each input corresponds to the decimal value between 0 and 9. The circuit outputs each representative equivalent binary of the compared digit of e included, for example, 2 as the output corresponding to input 0000 and 7 as the output to input 0001. If an input is greater or equal to 10 in decimal or non-binary all outputs are turned off and set to 0 to indicate an error. Each output is on only when the decimal place for the involved number matches the desired one, in order to make each valid input to correspond to a one-hot encoding. This circuit design utilizes logic equations that govern the outputs in similar structures laboratory reviewed to ensure accuracy and range compliance.

Digits Identifier Circuit for Euler's Number

PC/CP220 Project Phase III

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Logic Equation:

The positions corresponding to each digit are:

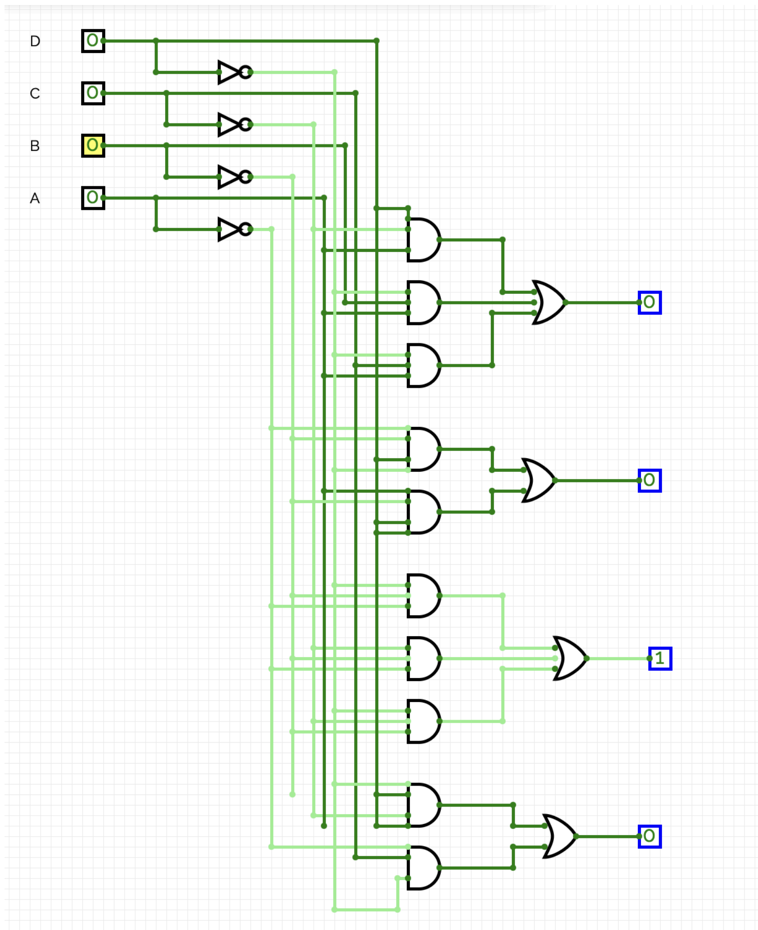
- Digit 2 is at positions 0,4,8
- Digit 1 is at positions 2 and 6
- Digit 7 is at position 1
- Digit 8 is at positions 3,5,7,9

For this Digit Identifier circuit, there are outputs O_0 to O_9 and equations for them are:

- O_0(Digit 2 at position 0,4,8):
  - O_0: $\overline{a3} \cdot \overline{a2} \cdot \overline{a1} \cdot \overline{a0} + \overline{a3} \cdot a_2 \cdot \overline{a1} \cdot \overline{a0} + a_3 \cdot \overline{a2} \cdot \overline{a1} \cdot \overline{a0}$
- O_1(Digit 1 at position 2 & 6):
  - O_1: $\overline{a3} \cdot \overline{a2} \cdot a_1 \cdot \overline{a0} + \overline{a3} \cdot a_2 \cdot a_1 \cdot \overline{a0}$
- $O_2 = O_3 = O_4 = O_5 = O_6$ (No positions corresponds to digit 2,3,4,5,6)
  - $O_2 = O_3 = O_4 = O_5 = O_6 : 0$
- O_7 (Digit 7 at position 1):
  - $O_7 = \overline{a3} \cdot \overline{a2} \cdot \overline{a1} \cdot a_0$
- O_8(Digit 8 at positions 3,5,7,9):
  - $O_8 = \overline{a3} \cdot \overline{a2} \cdot a_1 \cdot a_0 + \overline{a3} \cdot a_2 \cdot a_1 \cdot a_0 + a_3 \cdot \overline{a2} \cdot \overline{a1} \cdot a_0 + a_3 \cdot a_2 \cdot \overline{a1} \cdot a_0$
- O_9 (No position corresponds to digit 9):
  - $O_9 = 0$

Circuit:

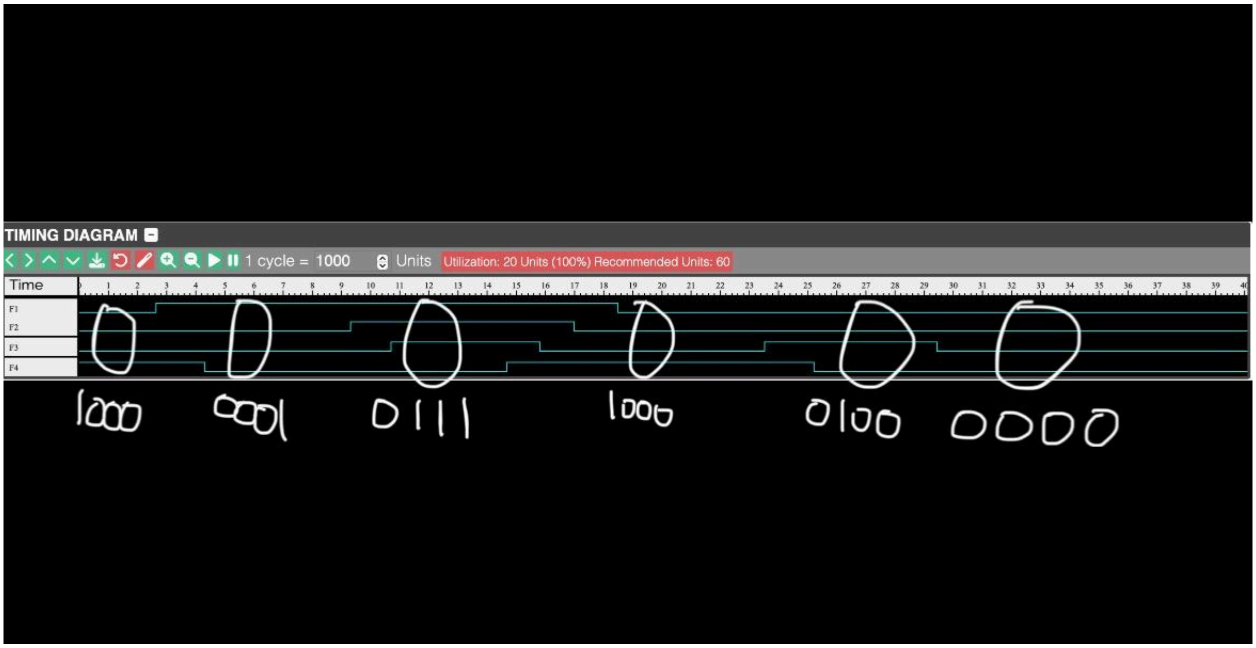
The Circuit was drawn using Circuit Verse and is shown in the following image:



Simulation output:

Following the simulations, the output shows that the result is true only for the correct input combinations, which have been circled:

- Binary = Decimal
    - 0010 = 2
    - 0111 = 7
    - 0001 = 1
    - 1000 = 8
    - 0010 = 2



- Binary = Decimal
    - 1000 = 8
    - 0001 = 1
    - 0111 = 7
    - 1000 = 8
    - 0100 = 4
    - 0000 = 0

Therefore this circuit is drawn correctly because the binary values equal to Euler's number 2.7182817840 (Decimal point cannot be included in binary thus it is omitted).

Parts List:

- Debugger Board for input
    - Only 1 should be required
- NOR Gates:
    - Roughly 4 would be required so each input also gets an NOR for it
- AND Gates:
    - Roughly 10 would be required so each digit in binary can be calculated accurately
- OR Gates:
    - Roughly 4 OR gates so each digit in binary can be calculated accurately
- Wire/ Connections
    - Multiple wires, specific number cannot be given as it can vary based on size

- 7 Segment Display for output:
    - Only 1 required